A Direct Quantitative Stability Theorem for Small-Amplitude Radial Graphs

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Abstract

This note proves the sharp Saint–Venant/Fraenkel estimate for a near-ball radial-graph class without using the BDV theorem as a black box. The coercive norm is not the full $H^{1/2}$ norm of the graph. The right coercive quantity for the main theorem is the L^2 size of the radial displacement, which controls the Fraenkel asymmetry. The proof is fully quantitative in principle: every constant below is obtained from elementary Taylor bounds, the Steklov spectrum of the ball, and a one-dimensional annular Poincare estimate.

1 Statement

Let

$$\Omega_R = \{r\theta : \theta \in \mathbb{S}^{n-1},\ 0 < r < R(\theta)\}, \qquad R = 1 + \varphi,$$

where R is smooth and positive. We normalize

$$|\Omega_R| = |B_1|, \qquad \int_{\Omega_R} x\,dx = 0.$$

Let u_Ω solve $-\Delta u_\Omega = 1$ in Ω_R, $u_\Omega = 0$ on $\partial\Omega_R$, and set $E(\Omega) = -\frac{1}{2}\int_\Omega u_\Omega$.

Theorem 1.1 (Direct radial-graph stability)**.** *For every $n \geq 2$ there are computable constants $\eta_0(n) > 0$, $\varepsilon_0(n) > 0$, and $c(n) > 0$ such that the following holds. If*

$$\|\varphi\|_{L^\infty(\mathbb{S}^{n-1})} \leq \eta_0(n), \qquad \|\varphi\|_{L^2(\mathbb{S}^{n-1})} \leq \varepsilon_0(n),$$

then

$$E(\Omega_R) - E(B_1) \geq c(n)\,\|\varphi\|^2_{L^2(\mathbb{S}^{n-1})}.$$

Consequently

$$E(\Omega_R) - E(B_1) \geq c_*(n)\,\mathcal{A}(\Omega_R)^2,$$

with another computable dimensional constant $c_*(n) > 0$.

In particular the theorem applies to any subclass with $\|\varphi\|_{H^{1/2}} \leq \varepsilon_0(n)$ and $\|\varphi\|_{L^\infty} \leq \eta_0(n)$, since $\|\varphi\|_{L^2} \leq \|\varphi\|_{H^{1/2}}$.

Remark 1.2. *The L^∞ amplitude assumption is not a regularity assumption: no slope, perimeter, curvature, C^1, or C^2 bound appears. It is used only to put the boundary in a thin annulus and to make the volume and barycenter constraints remove the constant and first spherical harmonics quantitatively.*

2 The exact energy identity

Let

$$u_0(x) = \frac{1-|x|^2}{2n}.$$

Define

$$z = u_{\Omega_R} - u_0 \qquad \text{in } \Omega_R.$$

Then z is harmonic in Ω_R, and on the boundary

$$z(R(\theta)\theta) = -u_0(R(\theta)\theta) = \frac{R(\theta)^2-1}{2n} =: h(\theta).$$

Lemma 2.1 (Exact graph identity)**.** *For every smooth positive radial graph R with $|\Omega_R| = |B_1|$,*

$$E(\Omega_R) - E(B_1) = \frac{1}{2}\int_{\Omega_R} |\nabla z|^2\,dx - \frac{1}{2n^2(n+2)}\int_{\mathbb{S}^{n-1}} \left(R^{n+2}-1\right) d\mathcal{H}^{n-1}.$$

Proof. Since $-\Delta u_0 = 1$, expanding

$$J_{\Omega_R}(v) = \frac{1}{2}\int_{\Omega_R} |\nabla v|^2 - \int_{\Omega_R} v$$

at u_0 gives

$$E(\Omega_R) = J_{\Omega_R}(u_0) + \int_{\partial\Omega_R} z\,\partial_\nu u_0\,d\mathcal{H}^{n-1} + \frac{1}{2}\int_{\Omega_R}|\nabla z|^2.$$

On $\partial\Omega_R$, $z = (R^2-1)/(2n)$ and $\partial_\nu u_0 = -x\cdot\nu/n$. The radial graph identity $(x\cdot\nu)\,d\mathcal{H}^{n-1} = R^n\,d\mathcal{H}^{n-1}_{\mathbb{S}^{n-1}}$ gives

$$\int_{\partial\Omega_R} z\,\partial_\nu u_0\,d\mathcal{H}^{n-1} = \frac{1}{2n^2}\int_{\mathbb{S}^{n-1}} (1-R^2)R^n\,d\mathcal{H}^{n-1}.$$

Also

$$J_{\Omega_R}(u_0) = -\frac{|B_1|}{2n} + \frac{n+1}{2n^2(n+2)}\int_{\mathbb{S}^{n-1}} R^{n+2}\,d\mathcal{H}^{n-1},$$

using $|\Omega_R| = |B_1|$. Subtracting the same formula with $R\equiv 1$ and using $\int_{\mathbb{S}^{n-1}} R^n\,d\mathcal{H}^{n-1} = |\mathbb{S}^{n-1}|$ yields the stated identity. □

3 Slope-free annular trace lower bound

Fix $0<\eta<1/4$, assume $1-\eta \le R \le 1+\eta$, and put $\rho = 1-\eta$. Write the spherical-harmonic expansion

$$h = \sum_{k\ge 0} h_k, \qquad h_k \in \mathcal{H}_k.$$

Lemma 3.1 (Annular Steklov lower bound)**.** *There is an explicit number $b_n(\eta) > 0$, with $b_n(\eta)\to 2$ as $\eta\downarrow 0$, such that every $w\in H^1(\Omega_R)$ whose radial trace on $r = R(\theta)$ is $h(\theta)$ satisfies*

$$\int_{\Omega_R} |\nabla w|^2\,dx \ge b_n(\eta)\sum_{k\ge 2} \|h_k\|^2_{L^2(\mathbb{S}^{n-1})}.$$

One admissible choice is

$$b_n(\eta) = \frac{2(1-\eta)^{n-2}\mu_n(\eta)}{2(1-\eta)^{n-2}+\mu_n(\eta)}, \qquad \mu_n(\eta) = \frac{(1-\eta)^{n-1}}{2\eta}.$$

Proof. Let $f(\theta) = w(\rho\theta)$. The energy inside B_ρ is bounded from below by the harmonic extension energy:

$$\int_{B_\rho} |\nabla w|^2 \ge \rho^{n-2} \sum_{k\ge1} k \, \|f_k\|^2_{L^2(\mathbb{S}^{n-1})} .$$

Along each ray, Cauchy's inequality gives

$$\int_\rho^{R(\theta)} |\partial_r w(r,\theta)|^2 r^{n-1}\, dr \ge \frac{|h(\theta)-f(\theta)|^2}{\int_\rho^{R(\theta)} r^{1-n}\, dr}.$$

Since $R(\theta) \le 1+\eta$, the denominator is at most

$$2\eta(1-\eta)^{1-n}.$$

Therefore the annular part of the energy is bounded below by

$$\mu_n(\eta)\, \|h-f\|^2_{L^2(\mathbb{S}^{n-1})}, \qquad \mu_n(\eta) = \frac{(1-\eta)^{n-1}}{2\eta}.$$

For each spherical harmonic degree $k \ge 1$, minimize

$$\rho^{n-2} k \, |f_k|^2 + \mu_n(\eta)|h_k - f_k|^2$$

over f_k. The minimum is

$$\frac{\rho^{n-2} k \, \mu_n(\eta)}{\rho^{n-2}k + \mu_n(\eta)} |h_k|^2.$$

This coefficient is increasing in k, so on $k \ge 2$ it is bounded below by the value at $k = 2$, which is exactly $b_n(\eta)$. ☐

4 Elementary consequences of the constraints

Lemma 4.1 (Removal of neutral modes)**.** *There are computable constants C_n and $\eta_1(n) > 0$ such that if $\|\varphi\|_{L^\infty} \le \eta_1(n)$, $|\Omega_R| = |B_1|$, and $\int_{\Omega_R} x\, dx = 0$, then*

$$\|P_0\varphi\|_{L^2} + \|P_1\varphi\|_{L^2} \le C_n \, \|\varphi\|^2_{L^2} .$$

Consequently, if also $\|\varphi\|_{L^2} \le \varepsilon_1(n)$, then

$$\|P_{\ge2}\varphi\|^2_{L^2} \ge \frac{15}{16} \, \|\varphi\|^2_{L^2} .$$

Proof. The volume condition is

$$0 = \int_{\mathbb{S}^{n-1}} \big((1+\varphi)^n - 1\big)\, d\mathcal{H}^{n-1} = n \int_{\mathbb{S}^{n-1}} \varphi \, d\mathcal{H}^{n-1} + O_n\Big(\int_{\mathbb{S}^{n-1}} \varphi^2 \, d\mathcal{H}^{n-1}\Big),$$

where the O_n-constant is explicit for $\|\varphi\|_{L^\infty} \le \eta_1(n)$. This gives $\|P_0\varphi\|_{L^2} \le C_n \, \|\varphi\|^2_{L^2}$.

The barycenter condition is

$$0 = \int_{\mathbb{S}^{n-1}} (1+\varphi)^{n+1}\theta \; d\mathcal{H}^{n-1} = (n+1) \int_{\mathbb{S}^{n-1}} \varphi\theta \; d\mathcal{H}^{n-1} + O_n\Big(\int_{\mathbb{S}^{n-1}} \varphi^2 \, d\mathcal{H}^{n-1}\Big),$$

which gives the P_1 estimate. The last claim follows by choosing $\varepsilon_1(n)$ so that the squared neutral-mode contribution is at most $\frac{1}{16} \, \|\varphi\|^2_{L^2}$. ☐

Lemma 4.2 (Moment and boundary data bounds)**.** *There are computable $\eta_2(n) > 0$ and C_n such that, if $\|\varphi\|_{L^\infty} \le \eta_2(n)$, then*

$$\int_{\mathbb{S}^{n-1}} \left(R^{n+2} - 1\right) d\mathcal{H}^{n-1} \le (n+2)(1 + C_n\eta_2) \|\varphi\|_{L^2}^2$$

and, for $h = (R^2 - 1)/(2n)$,

$$\|P_{\ge 2}h\|_{L^2}^2 \ge \frac{1 - C_n\eta_2}{n^2} \|P_{\ge 2}\varphi\|_{L^2}^2 - C_n \|\varphi\|_{L^2}^4 .$$

Proof. Taylor's formula gives

$$R^{n+2} - 1 = (n+2)\varphi + \frac{(n+2)(n+1)}{2}\varphi^2 + O_n(\eta_2\varphi^2),$$

and the volume expansion used in Lemma 4.1 gives

$$\int_{\mathbb{S}^{n-1}} \varphi \; d\mathcal{H}^{n-1} = -\frac{n-1}{2}\int_{\mathbb{S}^{n-1}} \varphi^2 \; d\mathcal{H}^{n-1} + O_n(\eta_2 \|\varphi\|_{L^2}^2).$$

Combining these two identities yields the moment estimate.

For the boundary datum,

$$h = \frac{\varphi}{n} + \frac{\varphi^2}{2n}.$$

Since $\left\|P_{\ge 2}(\varphi^2)\right\|_{L^2} \le \|\varphi\|_{L^\infty} \|\varphi\|_{L^2}$, the stated lower bound follows from $\|a+b\|^2 \ge (1 - \tau)\|a\|^2 - \tau^{-1}\|b\|^2$, with a fixed small τ, and then shrinking $\eta_2(n)$. □

5 Proof of the theorem

Choose $\eta_0(n) > 0$ so small that

$$b_n(\eta_0) \ge \frac{3}{2}, \qquad C_n\eta_0 \le \frac{1}{8},$$

where C_n is large enough for Lemmas 4.1 and 4.2. Then choose $\varepsilon_0(n) > 0$ so small that the quartic terms in Lemmas 4.1 and 4.2 are absorbed by $\frac{1}{16}\|\varphi\|_{L^2}^2$.

Apply Lemma 3.1 to $w = z$, then Lemmas 4.1 and 4.2:

$$\int_{\Omega_R} |\nabla z|^2 \ge \frac{3}{2}\|P_{\ge 2}h\|_{L^2}^2 \ge \frac{5}{4n^2}\|\varphi\|_{L^2}^2$$

after the above choices of η_0, ε_0. Also

$$\int_{\mathbb{S}^{n-1}} (R^{n+2} - 1)\, d\mathcal{H}^{n-1} \le \frac{9}{8}(n+2)\|\varphi\|_{L^2}^2 .$$

The exact identity in Lemma 2.1 therefore gives

$$\begin{aligned} E(\Omega_R) - E(B_1) &\ge \frac{1}{2}\frac{5}{4n^2}\|\varphi\|_{L^2}^2 - \frac{1}{2n^2(n+2)}\frac{9}{8}(n+2)\|\varphi\|_{L^2}^2 \\ &= \frac{1}{16n^2}\|\varphi\|_{L^2}^2 . \end{aligned}$$

This proves the first assertion, with the displayed constant after the explicit threshold choices above.

Finally,

$$|\Omega_R \Delta B_1| = \frac{1}{n}\int_{\mathbb{S}^{n-1}} |R^n - 1|\, d\mathcal{H}^{n-1} \le C_n \|\varphi\|_{L^1(\mathbb{S}^{n-1})} \le C_n \|\varphi\|_{L^2(\mathbb{S}^{n-1})} .$$

Since $\mathcal{A}(\Omega_R) \le |B_1|^{-1}|\Omega_R \Delta B_1|$, the Fraenkel estimate follows.

6 What this proves and what it does not prove

The proof is quantitative and does not invoke BDV's global theorem or BDV's nearly-spherical theorem. It proves the main theorem in a genuine small-amplitude radial-graph class with no slope or curvature control. The $H^{1/2}$ hypothesis is used only as a convenient way to guarantee small L^2; the coercive output is the L^2/Fraenkel scale.

The proof does not prove the false estimate

$$E(\Omega_R) - E(B_1) \gtrsim \|\varphi\|^2_{H^{1/2}}$$

without additional geometry. High-frequency capillary ripples still rule out that stronger statement. It also does not remove the need for an amplitude hypothesis: $H^{1/2}(\mathbb{S}^{n-1})$ alone does not put the graph in a thin annulus or even give an L^∞ radius bound.